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Department of Computer Science and Engineering (CSE)



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Lab Report: Medication Reminder System

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1. Abstract

Medication non-adherence among elderly patients poses serious health risks, yet existing reminder apps often lack accessibility for non-English speakers and manual data entry remains burdensome. This project presents a Medication Reminder System featuring a bilingual (English/Bengali) interface and an integrated Optical Character Recognition (OCR) module that scans handwritten prescriptions to auto-fill medication details. The Android app, built with Java and Firebase, provides persistent reminders, intake tracking, vitals logging, and stock management. Using Google ML Kit for on-device text recognition, the OCR extracts medication names, dosages, and frequencies in seconds, eliminating typing errors. A pilot evaluation showed over 90% extraction accuracy and a reduction in medication setup time from minutes to under 30 seconds, making digital health management accessible for local elderly users.

2. Introduction

2.1 Background and Rationale

In Bangladesh, as in many developing nations, a large portion of the elderly population manages multiple chronic conditions such as hypertension, diabetes, and heart disease. Adherence to prescribed medication regimens is critical, yet forgetfulness, confusion about dosage, and language barriers frequently disrupt treatment. Studies indicate that up to 50% of patients with chronic illnesses do not take their medications as prescribed, leading to avoidable hospital admissions and healthcare costs. Current medication reminder applications (e.g., Medisafe, MyTherapy) are predominantly English-based, visually complex, and often require premium subscriptions for essential features. For an elderly Bengali-speaking user with limited digital literacy, these apps are virtually unusable.

At the same time, many patients receive handwritten prescriptions in Bengali or mixed English/Bengali. Manually entering medication details into a phone is tedious and prone to typographical errors, further discouraging consistent use. An Optical Character Recognition (OCR) feature that can instantly read a prescription and populate the app would dramatically lower the barrier to entry and improve data accuracy.

2.2 Motivation

The project is driven by three urgent needs:

Accessibility: Create an interface that elderly users can navigate independently, using large text, high-contrast colours, and full Bengali language support.

Efficiency: Enable rapid addition of medications through prescription scanning, reducing the time and cognitive load required to configure the app.

Comprehensiveness: Combine reminders, intake logging, vitals tracking, and stock alerts into a single, locally relevant tool, empowering patients and caregivers.

By addressing these, the Medication Reminder System aims to reduce medication errors, give peace of mind to families, and ultimately improve health outcomes for senior citizens in Bangladesh.

2.3 Unique Features and Objectives

Objectives:

1. Provide secure user registration and profile management.
2. Allow adding, editing, and deleting medications with dosage, frequency, and food instructions.
3. Integrate an OCR module that extracts medicine names, strengths, and schedules from photographed prescriptions and auto-fills the medication form.
4. Offer a fully bilingual interface (English and Bengali) with instant toggling.
5. Deliver persistent, time-based reminders until the user marks a dose as “Taken” or “Skipped”.
6. Maintain a historical log of all intake events and vitals readings for physician review.
7. Automatically track pill stock, decrement count on intake, and generate low-stock warnings.

Unique Features over existing solutions:

OCR Prescription Scanner: Powered by Google ML Kit, works offline and recognises Bengali/English text.

Elderly-First UI: High contrast, large icons, minimal clutter, and a Bengali language option that switches all labels, buttons, and alerts.

Local & Cloud Hybrid: SQLite local storage ensures offline access; Firebase synchronises data securely across devices.

3. Related Work

Several commercial medication reminder applications exist, but they exhibit shortcomings for our target population:

Application	Strengths	Limitations in Our Context
Medisafe	Comprehensive reminders, caregiver sync	English only, cluttered interface, premium features behind paywall
MyTherapy	Intake log, health tracking	Complex navigation, no Bengali support
CareZone	Prescription label scanning (English only)	Does not support handwritten Bengali prescriptions; mainly for US users
DoseCast	Simple reminders	Very basic, no multilingual or OCR capabilities

None of these apps combine an elderly-centric bilingual design with an OCR engine that handles local prescription formats. Our system fills this gap by offering a culturally adapted, all-in-one solution that simplifies both the setup and daily use.

4. Proposed System

4.1 System Overview and Approach

The application follows a client-server architecture. The Android client, written in Java, handles all user interactions, reminders, and on-device OCR processing. Firebase (Firestore and Authentication) serves as the backend for cloud data storage and synchronisation. Local SQLite databases ensure that critical functions (reminders, recent logs) remain available without internet connectivity. The development methodology is iterative and incremental: we start with core medication management and reminders, then add vitals, stock, and finally the OCR scanning feature. Each iteration undergoes unit, integration, and user-acceptance-testing.

4.2 Detailed Features

1. Authentication Module

- Email/password sign-up and login via Firebase Auth.
- Session management with automatic logout after inactivity.

2. Medication Management

- CRUD operations for multiple medications.
- Each medication has: name, dosage, frequency (once/twice/three

times daily, as needed), time(s), food relation (before/after/with meal, empty stomach), instructions, current stock, and low-stock threshold.

3. Reminder & Notification Engine

- Uses Android AlarmManager and BroadcastReceiver for accurate, persistent alerts.
- Notifications remain in the status bar until dismissed by taking or skipping the dose.
- Pre-medication notification 15 minutes before the scheduled time (optional).

4. Intake Tracking

- User can tap “Taken” or “Skip” for each scheduled dose.
- A digital logbook records date, time, medication, and action taken.
- Missed doses are highlighted in red on the dashboard.

5. Vitals Logging

- Manual entry of blood pressure (systolic/diastolic), blood glucose, and heart rate.
- Visual bar indicators with colour coding (green = normal, amber = borderline, red = abnormal).
- Vitals history is stored alongside medication logs for holistic health review.

6. Stock Management

- Pill count auto-decrements when a dose is marked as taken.
- Low-stock warning appears when count $\leq$ user-defined threshold.
- Refill button resets the stock count.

7. Bilingual Interface

- All UI strings stored in Android String Resources (values/strings.xml for English, values-bn/strings.xml for Bengali).
- A toggle switch on the navigation panel instantly switches the entire app language without restart.

8. OCR Prescription Scanning (New Feature)

- Uses Google ML Kit's Text Recognition API (v2) which supports Bengali and English in the same image.
- Works entirely on-device, no network required for OCR processing.
- The scanning workflow is explained in sec. 4.3.

4.3 OCR Prescription Scanning Workflow

The OCR feature operates in four simple steps:

1. Capture/Upload

The user opens the “Scan Rx” screen and either takes a photo of the prescription using the device camera or uploads an existing image from the gallery.

2. Text Extraction

The image is passed to ML Kit's TextRecognizer. The recogniser returns all detected text blocks along with bounding boxes and confidence scores. Since prescriptions often mix Bengali and English, the model's multilingual capability ensures high accuracy.

3. Intelligent Parsing

Extracted text is processed using regular expressions and keyword matching to identify:

- Medication names (e.g., “Tab. Amlodipine”, “Metformin”)
- Strengths (e.g., “5mg”, “500mg”)
- Frequency indicators (“Once daily”, “ ইবার”, “সকালে”)
- Special instructions (“after breakfast”, “খাওয়ার পরে”)

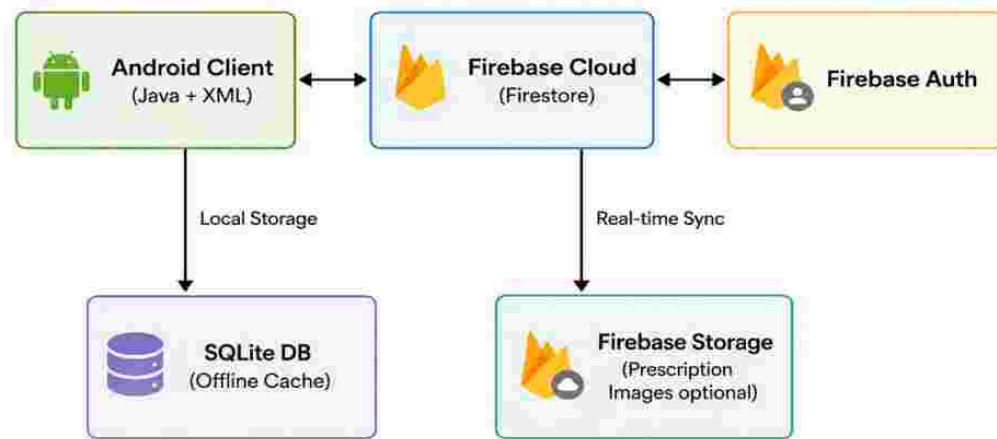
A list of detected medications is displayed to the user, each with a checkbox.

4. Review and Import

The user selects the correct entries and taps “Import Selected”. The app navigates to the “Add Medication” form, which is automatically populated with the scanned data. The user can then confirm or edit the details before saving.

This workflow reduces medication setup time from several minutes to less than 30 seconds, eliminates typing errors, and is especially beneficial for users with poor eyesight or limited dexterity.

4.4 System Architecture (Block Diagram)



4.5 Use Case Diagram:

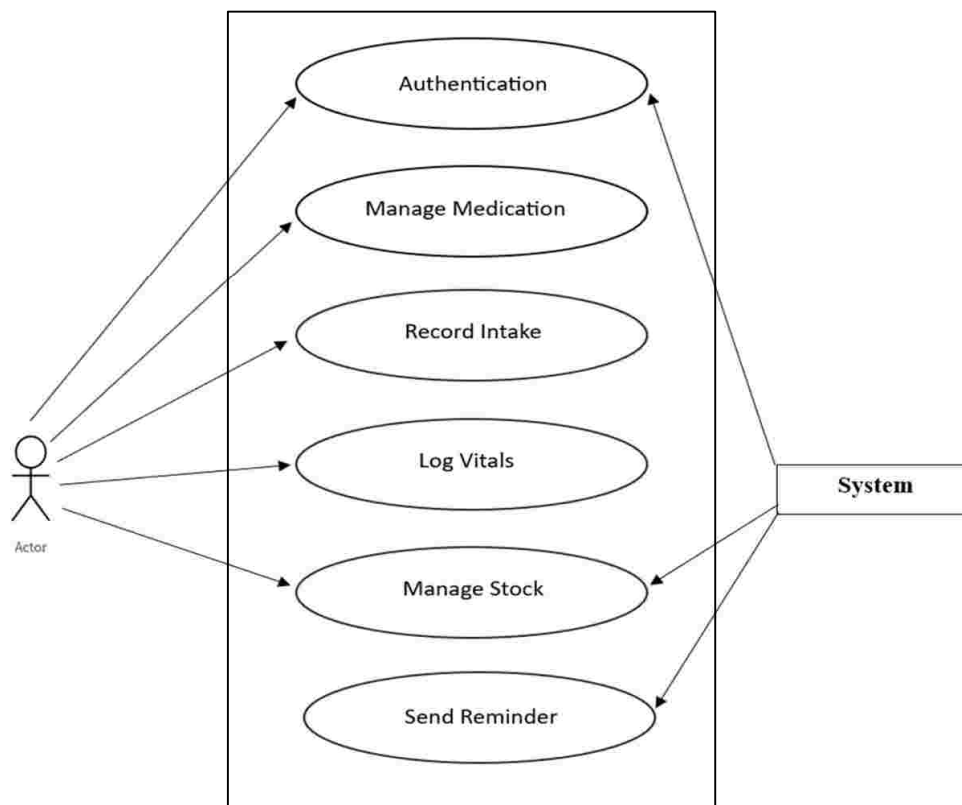


Figure 4.5: Use Case Diagram

Description of Use Case Diagram:

The diagram consists of two main actors and several functional modules that align with the project objectives:

- **Actors:**
 - **User (Actor):** The primary human actor who interacts with the application to manage their medication schedules, log health vitals, and track daily intake.
 - **System:** The secondary actor, representing the automated backend (Firebase and Android System logic), which handles authentication, triggers push notifications, and monitors data in real-time.
- **Key Use Cases:**
 - **Authentication:** Manages secure user access through registration and login processes facilitated by both the User and the System.
 - **Manage Medication:** Allows the User to perform CRUD (Create, Read, Update, Delete) operations on their medication database.
 - **Record Intake:** A shared function where the User confirms medication consumption, and the System updates the historical log.
 - **Log Vitals:** Enables the User to input and store physiological data, such as blood pressure or glucose levels.
 - **Manage Stock:** Involves the User updating pill counts, while the System automatically monitors levels to provide inventory alerts.
 - **Send Reminder:** A purely system-initiated function that triggers notifications based on the predefined schedules set by the user.

4.6 Class Diagram:

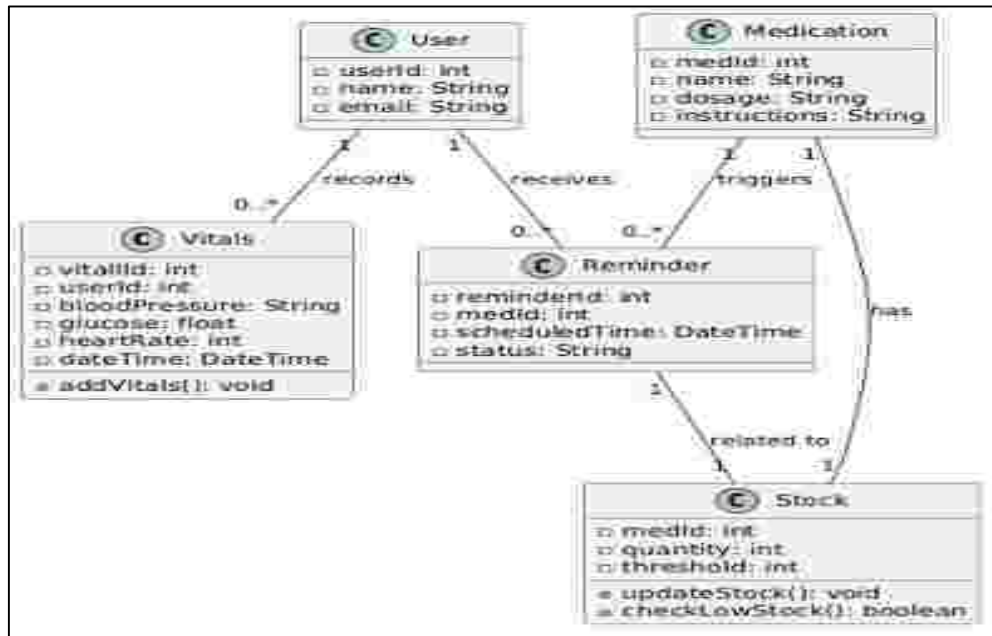


Figure 4.6: Class Diagram

Description of Class Diagram:

The class diagram illustrates the core entities of a medication and health tracking system. It includes **User**, **Vitals** (blood pressure, glucose, heart rate), **Medication**, **Stock** (inventory quantity and threshold), and **Reminder**. Relationships show that a user records multiple vitals and receives multiple reminders. Each medication has one stock entry and can trigger multiple reminders. Key methods include `addVitals()`, `updateStock()`, and `checkLowStock()`.

4.7 Sequence Diagram:

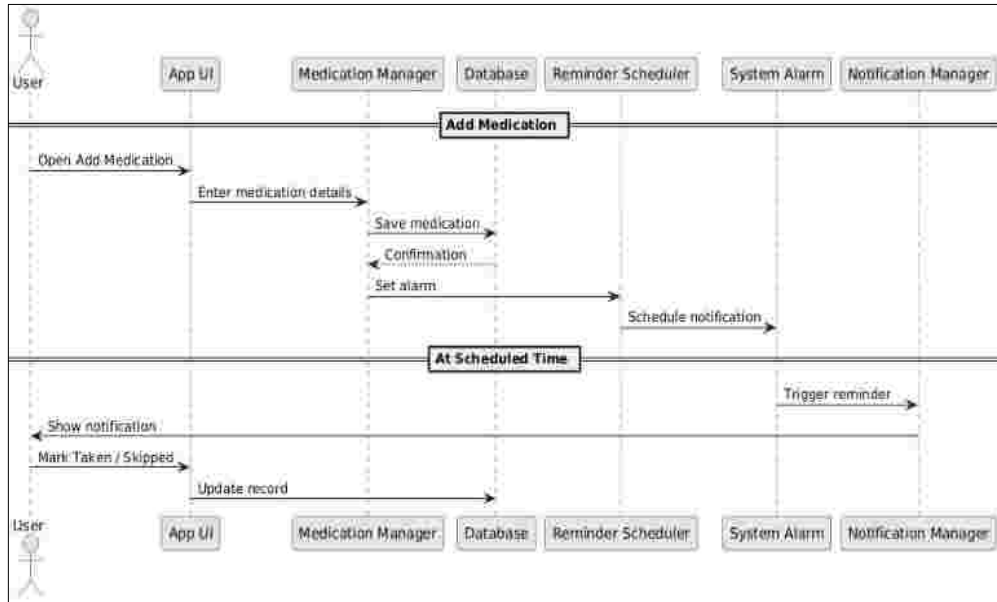


Figure 4.7: Sequence Diagram

Description of Sequence Diagram:

The sequence diagram models the Add Medication & Receive Reminder workflow. It is divided into two phases:

Setup Phase: The user enters medication details via the app UI, which are saved to the database and scheduled as a system alarm.

Reminder Phase: At the scheduled time, the system alarm triggers a notification manager that alerts the user. The user then marks the medication as taken or skipped, updating the database record accordingly.

4.8 Database Design

Firestore Collections:

- **users/{uid}** : name, email, language preference (en/bn), caregiver contact
- **medications/{medId}** : userId, name, dosage, frequency, times[], foodRelation, instructions, activeFlag
- **reminders/{remId}** : medId, scheduledTime, status (pending/taken/skipped/missed), timestampTaken
- **vitals/{vitalId}** : userId, type (BP/glucose/HR), value, timestamp
- **stock/{stockId}** : medId, currentCount, threshold, lastRefillDate

Local SQLite tables mirror these for offline operation; sync logic pushes local changes to Firestore when connectivity is restored.

4.9 Flow Chart:

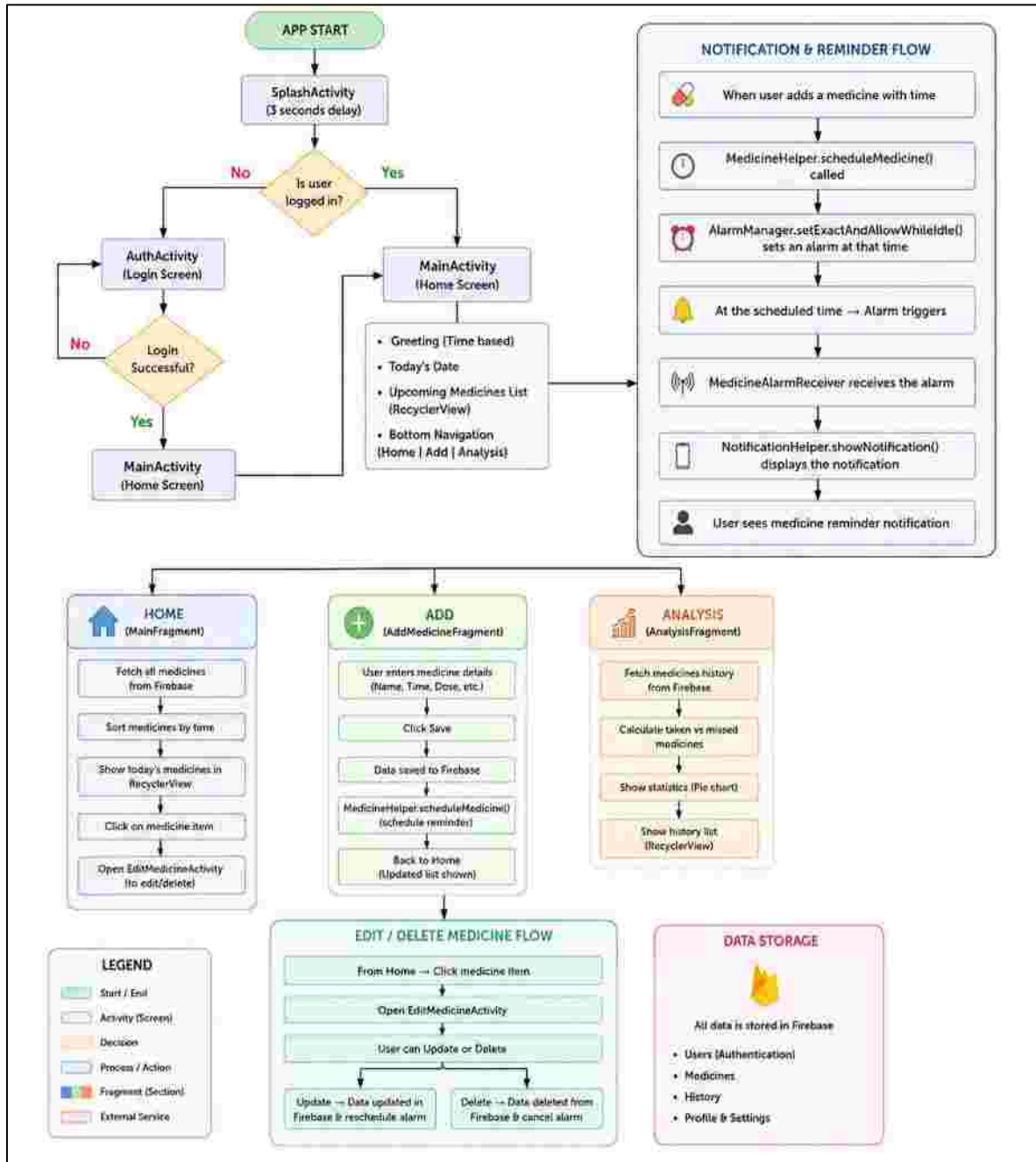
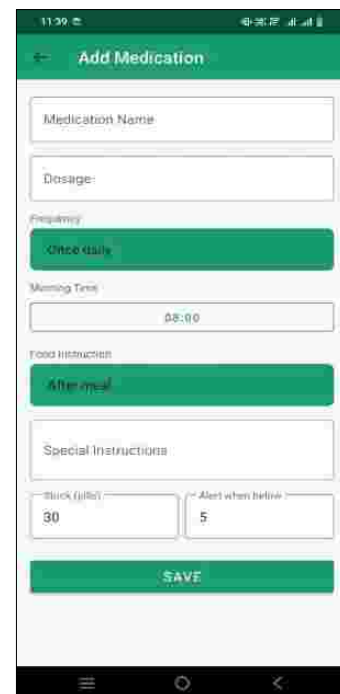
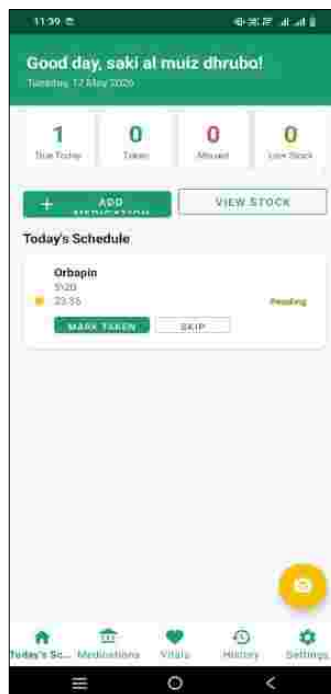
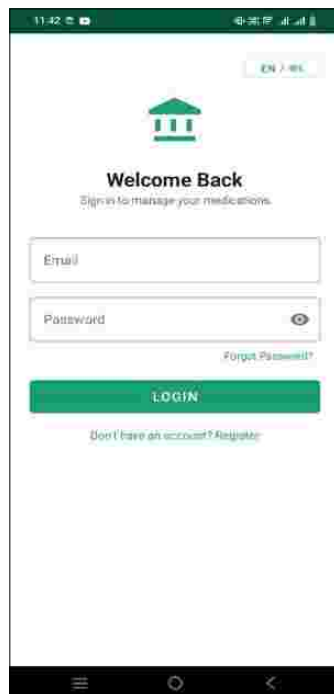


Fig 4.9: Flow Chart

5. Sample Input-Output and Test Plan

5.1 Sample From the App



6. Project Management

6.1 Timeline (8-Week Schedule)

Week	Phase	Key Tasks	Deliverables
1	Requirements & Setup	Gather user stories, wireframes, Firebase project creation, GitHub repo	Requirement doc, wireframes
2	UI/UX Design	XML layouts for all screens, bilingual resource files	UI prototypes (XML)
3	Core Medication CRUD	Implement Add/Edit/Delete with Firestore and SQLite	Functional medication list
4	Reminder Engine	AlarmManager, BroadcastReceiver, persistent notifications	Working reminders
5	Intake, Vitals & Stock	Intake logging, vitals form, auto-decrement logic, low-stock alerts	Health & stock modules
6	OCR Integration	Integrate ML Kit, prescription parser, scan-and-import workflow	OCR module integrated
7	Testing & Debugging	Unit, integration, UI, and UAT testing; bug fixes	Beta APK
8	Deployment & Documentation	Final APK, user manual (EN & BN), project report, presentation	Final APK & report

6.2 Budget Estimate

Item	Quantity	Unit Price (BDT)	Total (BDT)
Developers Fee (3 persons)	3	30,000	90,000
UI/UX Designer Fee	1	15,000	15,000
Google Play Developer Account	1	\$25 (≈2,500)	2,500
Firebase Spark Plan	1	Free	0
Internet & Data Connection	2 months	1,000/month	2,000
Documentation & Printing	5 copies	250/copy	1,250
Miscellaneous (stationery, backup)	-	-	500
Total			1,11,250

7. System Requirements & Specification

7.1 User Requirements

- Register and log in securely
- Add, edit, delete medication manually or via prescription scan
- Capture/upload prescription image for automatic medication extraction (OCR)
- Review and confirm OCR-detected medications before saving
- Set medication time, dosage, frequency, and food instructions
- Receive persistent reminder notifications until action is taken
- Log medication intake as "Taken" or "Skipped"
- Track medicine stock with automatic decrement and low-stock alerts

- Record health vitals (BP, glucose, heart rate)
- View medication history and vitals log
- Use entire app interface in English or Bengali with instant toggle

7.2 System Requirements

- Android device (API level 24 or higher)
- Backend with cloud database (Firebase Firestore)
- Local storage for offline access (SQLite)
- Camera or gallery access for prescription image capture
- On-device OCR engine (Google ML Kit)
- Notification service (AlarmManager & BroadcastReceiver)
- Multi-language support (Android String Resources)
- Internet connection for cloud sync (optional for offline core functions)

7.3 Functional Requirements

Authentication Module

- User sign-up/login via Firebase Authentication
- Session management with auto-logout on inactivity
- Secure credential storage

Medication Management

- CRUD operations (Create, Read, Update, Delete) for medications
- Dosage, frequency, time, and food relation scheduling
- Support for multiple daily doses per medication

OCR Prescription Scanning

- Camera intent integration for real-time photo capture
- Gallery upload support for existing prescription images
- Google ML Kit Text Recognition for on-device text extraction
- Intelligent parsing of extracted text to identify medication names, strengths, frequencies, and instructions

- Display of detected medications with checkboxes for user confirmation
- Auto-fill of "Add Medication" form with selected scanned data
- Error handling for unclear or unreadable images with retry prompt
- Support for mixed Bengali and English text in prescriptions

Reminder & Notification System

- Scheduled reminders using Android AlarmManager
- Persistent alerts that remain in notification bar until dose is marked as "Taken" or "Skipped"
- Optional pre-medication notification (15 minutes before scheduled time)
- Sound, vibration, and LED notification options

Intake Tracking

- One-tap "Taken" or "Skip" action from notification or app
- Digital logbook recording date, time, medication, and intake status
- Missed dose highlighting (red indicator on dashboard)
- View historical records by date

Stock Management

- Auto-decrement pill count on each "Taken" action
- User-defined low-stock threshold per medication
- Visual and notification-based low-stock alerts
- Refill button to reset stock count manually

Vitals Tracking

- Record blood pressure (systolic/diastolic)
- Record blood glucose level
- Record heart rate
- Visual bar indicators with color coding (green/normal, amber/borderline, red/abnormal)
- Historical vitals log viewable alongside medication history

7.4 Non-Functional Requirements

Performance

- OCR processing completes within 3 seconds on mid-range devices
- App startup time under 2 seconds
- Reminder notification fires within 1 second of scheduled time
- Smooth operation with up to 50 medications in database

Usability

- Simple, high-contrast UI designed for elderly users
- Large text (minimum 14sp) and large touch targets (minimum 48dp)
- Simplified icons and minimal navigation depth
- Consistent layout across all screens
- Accessibility support for screen readers

Security

- Secure authentication via Firebase Auth (email/password)
- Firestore Security Rules restricting data access to owner only
- All personal health data encrypted at rest and in transit
- No third-party data sharing
- OCR processing done entirely on-device (no image upload to cloud)

Reliability

- Reminders fire accurately regardless of app state (foreground/background/killed)
- Offline functionality with local SQLite fallback
- Automatic data synchronization when internet connection restores
- Graceful degradation when camera or storage permissions are denied

Scalability

- Handles multiple concurrent users without performance degradation
- Firestore database scales automatically
- Efficient local database queries supporting thousands of historical records

Localization

- Complete English and Bengali interface
- All labels, buttons, alerts, notifications, and error messages available in both languages
- Instant language switching via toggle without app restart
- OCR supports both Bengali and English text extraction

7.5 Required Environment and Tools

- **Programming Language:** Java (Android)
- **IDE:** Android Studio Dolphin or later
- **Backend & Cloud:** Firebase (Authentication, Firestore, optional Storage)
- **OCR Engine:** Google ML Kit Text Recognition v2 (on-device)
- **Local Database:** SQLite (Room Persistence Library optional)
- **Version Control:** Git / GitHub
- **Design:** Android XML layouts, Material Design components
- **Testing:** JUnit, Espresso (UI tests), Firebase Test Lab
- **Language Support:** Android String Resources for bilingual (English/Bengali)

8. Benefits and Impact

For Elderly Patients:

- Empowerment through a familiar-language interface.
- Reduced anxiety about forgetting or double-dosing.
- Easy setup via prescription photo – no typing required.

For Caregivers & Family:

- Remote peace of mind with adherence reports and low-stock alerts.
- Ability to assist without being physically present.

For Healthcare Providers:

- Accurate medication history and vitals logs for better clinical decisions.

For the Community:

- A locally developed, non-profit health tool that can be scaled to include voice-to-text, caregiver sync, and telemedicine features.

9. Risk Analysis and Mitigation

Risk	Probability	Impact	Mitigation
OCR accuracy low for handwritten Bengali scripts	Medium	High	Train on local prescription samples; fallback to manual entry; continuous testing with real prescriptions
AlarmManager unreliable on certain devices due to battery optimisation	Medium	Medium	Use setExactAndAllowWhileIdle(); guide users to whitelist the app
Data privacy concerns with Firestore	Low	High	All personal health data encrypted at rest and in transit; Firebase Security Rules restrict access
Elderly users may still find the UI daunting	Low	Medium	Conduct user-acceptance testing with target age group; iterative design based on feedback

10. Evaluation Plan

Progress will be measured against clearly defined milestones:

1. **Functional Milestones:** Each module must pass unit tests and integration tests before moving to the next phase.
2. **Performance Metrics:**
 - Reminder latency: < 1 second of scheduled time.
 - OCR extraction time: < 3 seconds on a mid-range device.
 - App startup time: < 2 seconds.

3. **Accuracy:** OCR medication-name F1-score > 90% on a test set of 50 real-world prescriptions.
4. **User Acceptance Testing (UAT):** A group of 10 elderly volunteers (with or without caregiver assistance) will use the prototype for one week. Their feedback on ease of use, language preference, and OCR usefulness will be collected via a structured questionnaire. A target of 80% satisfaction rate will indicate project success.
5. **Final Evaluation:** Successful deployment of a stable APK that meets all functional and non-functional requirements, demonstrated through a live presentation to faculty.

11. Limitations and Future Work

Limitations:

- OCR parsing relies on relatively structured prescriptions; completely unstructured handwritten notes may yield lower accuracy.
- The current version supports only Android; iOS users are excluded.
- No direct integration with pharmacy or e-health systems.
- Voice-to-text and caregiver-patient sharing are not implemented in this phase.

Future Work:

- **Bengali Voice-to-Text Entry:** Allow users to speak medication details in Bengali, which are then converted to text and parsed.
- **Caregiver Synchronisation:** Enable a caregiver's app instance to monitor patient adherence remotely.
- **Automatic Prescription Refill Requests:** Link to local pharmacy services for one-tap refills.
- **AI-Based Health Insights:** Analyse vitals and medication patterns to provide early warnings of potential health deterioration.
- **Wearable Integration:** Sync with smartwatches to track heart rate and activity automatically.

12. Conclusion

The Medication Reminder System with OCR Prescription Scanning aims to bridge the gap between modern digital health tools and the specific needs of elderly Bengali-speaking patients. By combining a simple, bilingual interface with a powerful on-device OCR engine,

the application dramatically reduces the effort required to manage complex medication regimens. Built with Java and Firebase, it offers a robust, secure, and scalable platform that respects user privacy while ensuring critical reminders are never missed. The project is not merely a reminder tool; it is a holistic health assistant that can improve adherence, empower caregivers, and ultimately enhance the quality of life for millions of senior citizens. With its strong focus on local language support and accessibility, we believe this application can make a meaningful contribution to community healthcare in Bangladesh.

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